Derivatives of Inverse and Inverse-Trigonometric Functions Answer Key

AP Calculus AB/BC — Differentiation

Quick Answers

1. $\frac{3}{9x^2+1}$
2. $\frac{2x}{\sqrt{1-x^4}}$
3. 1, $\frac{1}{5}$
4. $\frac{x}{x^2+1} + \operatorname{atan}(x)$
5. $-\frac{2}{\sqrt{1-4x^2}}$
6. $-\frac{2x}{(x^2+1)^2}$
7. $\frac{1}{3}$
8. $\left(\operatorname{atan}(x) + \frac{1}{x^2+1}\right) e^x$
9. $\frac{1}{2\sqrt{x}(x+1)}$
10. $-\frac{16x}{(4x^2+1)^2}$

Every solution names the inner function u (or the rule being combined) before it differentiates. On the AP exam the chain-rule factor u' is the mark most often lost, so it is written explicitly in each step.

1. Differentiate $y = \arctan(3x)$.

Solution: $u = 3x$, so $u' = 3$, and $\dfrac{d}{dx}\arctan(u) = \dfrac{u'}{1+u^2}$:

$$y' = \frac{3}{1+(3x)^2}$$

$$\boxed{y' = \frac{3}{1+9x^2}}$$

2. Differentiate $y = \arcsin(x^2)$.

Solution: $u = x^2$, so $u' = 2x$, and $\dfrac{d}{dx}\arcsin(u) = \dfrac{u'}{\sqrt{1-u^2}}$:

$$y' = \frac{2x}{\sqrt{1-(x^2)^2}}$$

$$\boxed{y' = \frac{2x}{\sqrt{1-x^4}}}$$

Common error: writing $\sqrt{1-x^2}$ in the denominator — the square is applied to u, not to x.

3. $f(x) = x^3 + 2x + 1$; find b with $f(b) = 4$, then $\left(f^{-1}\right)'(4)$.

(a) Solution: Solve $x^3+2x+1 = 4$, i.e. $x^3+2x-3 = 0$. Factoring gives $(x-1)(x^2+x+3) = 0$, and the quadratic has no real root, so the only real solution is $\boxed{b = 1}$

(b) Solution: $f'(x) = 3x^2+2$, so $f'(1) = 5$. The inverse-function rule $\left(f^{-1}\right)'(a) = \dfrac{1}{f'(b)}$ with $b = 1$ gives $\boxed{\left(f^{-1}\right)'(4) = \frac{1}{5}}$

Note: the reciprocal is of f' evaluated at the *input* b, never at the output a; $1/f'(4) = 1/50$ is the classic wrong answer.

4. Differentiate $y = x\arctan(x)$.
Solution: Product rule with $f = x$, $g = \arctan x$:

$$y' = (1)\arctan(x) + x \cdot \frac{1}{1+x^2}$$

$$\boxed{y' = \arctan(x) + \frac{x}{1+x^2}}$$

5. Differentiate $y = \arccos(2x)$.
Solution: $u = 2x$, $u' = 2$, and $\dfrac{d}{dx}\arccos(u) = \dfrac{-u'}{\sqrt{1-u^2}}$ — the arccos derivative is the negative of the arcsin one:

$$y' = \frac{-2}{\sqrt{1-(2x)^2}}$$

$$\boxed{y' = \frac{-2}{\sqrt{1-4x^2}}}$$

6. Second derivative of $y = \arctan(x)$.
Solution: $y' = \dfrac{1}{1+x^2} = (1+x^2)^{-1}$. Differentiate again with the chain rule (inner $1+x^2$, derivative $2x$):

$$y'' = -1(1+x^2)^{-2}(2x)$$

$$\boxed{y'' = \frac{-2x}{(1+x^2)^2}}$$

Reading: y'' is positive for $x < 0$ and negative for $x > 0$, matching the S-shape of arctan with its inflection at the origin.

7. $g(x) = 2x + \sin(x)$; compute $\left(g^{-1}\right)'(0)$.
Solution: First the input: $g(0) = 0 + \sin 0 = 0$, so $b = 0$. Then $g'(x) = 2 + \cos(x)$, giving $g'(0) = 2 + 1 = 3$, and $\boxed{\left(g^{-1}\right)'(0) = \frac{1}{3}}$

Since $g'(x) = 2 + \cos x \geq 1 > 0$, g is strictly increasing everywhere, which is what guarantees the inverse exists.

8. Differentiate $y = e^x\arctan(x)$.
Solution: Product rule; the derivative of e^x is itself.

$$y' = e^x\arctan(x) + e^x \cdot \frac{1}{1+x^2}$$

$$\boxed{y' = e^x\left(\arctan(x) + \frac{1}{1+x^2}\right)}$$

9. Differentiate $y = \arctan(\sqrt{x})$.

Solution: $u = \sqrt{x} = x^{1/2}$, so $u' = \dfrac{1}{2\sqrt{x}}$ and $u^2 = x$:

$$y' = \frac{1}{1+x} \cdot \frac{1}{2\sqrt{x}}$$

$$\boxed{y' = \frac{1}{2\sqrt{x}\,(1+x)}}$$

Why the denominator collapses: $(\sqrt{x})^2 = x$ exactly, so no radical survives in the first factor.

10. Challenge: second derivative of $y = \arctan(2x)$.

Solution: First derivative by the chain rule ($u = 2x$, $u' = 2$): $y' = \dfrac{2}{1+4x^2} = 2(1 + 4x^2)^{-1}$. Differentiate again, chain rule on $1 + 4x^2$ (derivative $8x$):

$$y'' = 2 \cdot (-1)(1 + 4x^2)^{-2}(8x)$$

$$\boxed{y'' = \frac{-16x}{(1+4x^2)^2}}$$

Check the structure: y'' vanishes only at $x = 0$, the inflection point of $\arctan(2x)$, and the outer constant 2 times the inner $8x$ is where the 16 comes from.